A. Match A, B, C and D with diffusion model, flow model, VAE and GAN.

Model	Efficient	Sample quality	Well-behaved latent space	Disentangled latent space	Efficient likelihood
A	✓	✓	✓	?	n/a
B	✓	✗	✓	?	✗
C	✓	✗	✓	?	✓
D	✗	✓	✗	✗	✗

Model Characteristics Analysis:
Efficient: Models A, B, and C are efficient, while Model D is not.
Sample quality: Models A and D have good sample quality, while B and C do not.
Well-behaved latent space: Models A, B, and C have well-behaved latent spaces, while D does not.
Efficient likelihood: Model C has efficient likelihood, Models B and D do not, and it's "n/a" for Model A.
Matching models:
Model A → Diffusion Model
Model B → VAE (Variational Autoencoder)
Model C → Flow Model
Model D → GAN (Generative Adversarial Network)
Therefore:
A = Diffusion Model
B = VAE (Variational Autoencoder)
C = Flow Model
D = GAN (Generative Adversarial Network)

What normalization is needed for a function f(x)= exp[-(x-4)2/16] to be used as the partition function in an energy-based generative model?

The function is:

f(x)=exp[−(x−4)216]f(x) = \exp\left[-\frac{(x-4)^2}{16}\right]f(x)=exp[−16(x−4)2]

In energy-based models, the probability distribution is defined as:

p(x)=exp(−E(x))Zp(x) = \frac{\exp(-E(x))}{Z}p(x)=Zexp(−E(x))

where E(x)E(x)E(x) is the energy function and ZZZ is the partition function that ensures the probabilities integrate to 1:

Z=∫exp(−E(x))dxZ = \int \exp(-E(x)) dxZ=∫exp(−E(x))dx

Looking at our function f(x)f(x)f(x), we can identify:

f(x)=exp(−E(x))f(x) = \exp(-E(x))f(x)=exp(−E(x))

where E(x)=(x−4)216E(x) = \frac{(x-4)^2}{16}E(x)=16(x−4)2

To use this as a valid probability distribution, we need to normalize it by calculating the partition function:

Z=∫−∞∞exp[−(x−4)216]dxZ = \int_{-\infty}^{\infty} \exp\left[-\frac{(x-4)^2}{16}\right] dxZ=∫−∞∞exp[−16(x−4)2]dx

This is a Gaussian integral. Note that (x−4)216\frac{(x-4)^2}{16}16(x−4)2 can be written as (x−4)22σ2\frac{(x-4)^2}{2\sigma^2}2σ2(x−4)2 where σ2=8\sigma^2 = 8σ2=8.

For a Gaussian distribution with mean μ\muμ and variance σ2\sigma^2σ2, we know:

∫−∞∞exp[−(x−μ)22σ2]dx=σ2π\int_{-\infty}^{\infty} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] dx = \sigma\sqrt{2\pi}∫−∞∞exp[−2σ2(x−μ)2]dx=σ2π

Applying this formula with μ=4\mu = 4μ=4 and σ=22\sigma = 2\sqrt{2}σ=22:

Z=222π=24π=4πZ = 2\sqrt{2}\sqrt{2\pi} = 2\sqrt{4\pi} = 4\sqrt{\pi}Z=222π=24π=4π

Therefore, the normalized probability distribution would be:

p(x)=f(x)Z=exp[−(x−4)216]4πp(x) = \frac{f(x)}{Z} = \frac{\exp\left[-\frac{(x-4)^2}{16}\right]}{4\sqrt{\pi}}p(x)=Zf(x)=4πexp[−16(x−4)2]

This gives us a proper probability distribution that integrates to 1, which is necessary for an energy-based generative model.

The normalization factor value is $1/4\sqrt{\pi}$

*Let μ_{data} = **N**(3,3) and μ_g = **N**(5,2) be the training data distribution and generated data distribution, respectively. What is the optimal discriminator function for a GAN?*

- Data distribution: μdata = N(3,3) [Normal distribution with mean 3 and variance 3]
- Generated distribution: μg = N(5,2) [Normal distribution with mean 5 and variance 2]

In the original GAN formulation by Goodfellow et al., the optimal discriminator D*(x) for fixed distributions μdata and μg is given by:

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x)+p_g(x)}$$

Where:

- p_data(x) is the probability density function of the data distribution μdata
- p_g(x) is the probability density function of the generated distribution μg

For normal distributions:

$$p(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Let's calculate the densities:

For μdata = N(3,3):

$$p_{data}(x) = \frac{1}{\sqrt{2\pi\cdot 3}} \exp\left(-\frac{(x-3)^2}{2\cdot 3}\right) = \frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(x-3)^2}{6}\right)$$

For μg = N(5,2):

$$p_g(x) = \frac{1}{\sqrt{2\pi\cdot 2}} \exp\left(-\frac{(x-5)^2}{2\cdot 2}\right) = \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(x-5)^2}{4}\right)$$

Now I can substitute these into the optimal discriminator formula:

$$D^*(x) = \frac{\frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(x-3)^2}{6}\right)}{\frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(x-3)^2}{6}\right) + \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(x-5)^2}{4}\right)}$$

Now I can substitute these into the optimal discriminator formula:

$$D^*(x) = \frac{\frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(x-3)^2}{6}\right)}{\frac{1}{\sqrt{6\pi}} \exp\left(-\frac{(x-3)^2}{6}\right) + \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(x-5)^2}{4}\right)}$$

Let's simplify by multiplying top and bottom by $\sqrt{6\pi}$:

$$D^*(x) = \frac{\exp\left(-\frac{(x-3)^2}{6}\right)}{\exp\left(-\frac{(x-3)^2}{6}\right) + \frac{\sqrt{6\pi}}{\sqrt{4\pi}} \exp\left(-\frac{(x-5)^2}{4}\right)}$$

$$D^*(x) = \frac{\exp\left(-\frac{(x-3)^2}{6}\right)}{\exp\left(-\frac{(x-3)^2}{6}\right) + \sqrt{\frac{6}{4}} \exp\left(-\frac{(x-5)^2}{4}\right)}$$

$$D^*(x) = \frac{\exp\left(-\frac{(x-3)^2}{6}\right)}{\exp\left(-\frac{(x-3)^2}{6}\right) + \sqrt{\frac{3}{2}} \exp\left(-\frac{(x-5)^2}{4}\right)}$$

This is the mathematical expression for the optimal discriminator function for this specific GAN with the given data and generator distributions.

To express this more compactly, we can also write:

$$D^*(x) = \frac{1}{1 + \sqrt{\frac{3}{2}} \exp\left(-\frac{(x-5)^2}{4} + \frac{(x-3)^2}{6}\right)}$$

This is the optimal discriminator for distinguishing between the data distribution N(3,3) and the generated distribution N(5,2).

Assume a hypothetical generative model that has been trained with (0, 0), (0, 1), (1, 0) and (1, 1). (0, 0) and (0, 1) are in class 1, and (1, 0) and (1, 1) in class 2. The generated four data points are (0.9, 0.2), (0.2, 0.3), (0.7, 0.7), and (0.1,0.1). Calculate

i) the Inception Score (IS) and

ii) Frechet Inception distance score (FID).

Clearly mention all your assumptions. Comment on the quality of the generated data based on these scores.

Training Data:

- Class 1: (0, 0), (0, 1)

- Class 2: (1, 0), (1, 1)

Generated Data:

- (0.9, 0.2), (0.2, 0.3), (0.7, 0.7), (0.1, 0.1)

Assumptions

1. We need a classifier to calculate inception scores. I'll use a simple distance-based classifier where each generated point is assigned to the class of its nearest training point.
2. For FID, I'll treat each class as a separate distribution and calculate the mean and covariance for both real and generated data within each class.
3. We'll consider the 2D points as features (similar to embeddings from an Inception network).
4. I'll assume uniform marginal distribution for generated samples.

i) Calculating Inception Score (IS)

First, I need to classify each generated point:

Nearest neighbor classification:

- (0.9, 0.2) - Closest to (1, 0) → Class 2
- (0.2, 0.3) - Closest to (0, 0) → Class 1
- (0.7, 0.7) - Closest to (1, 1) → Class 2
- (0.1, 0.1) - Closest to (0, 0) → Class 1

So we have 2 points classified as Class 1 and 2 points as Class 2.

The Inception Score is calculated as:
$IS = \exp(E[KL(p(y|x) \,||\, p(y))])$

Where:

- $p(y|x)$ is the conditional class distribution for each generated sample
- $p(y)$ is the marginal class distribution across all generated samples

For each generated point, $p(y|x)$ is a one-hot vector:

- For points in Class 1: $p(y|x) = [1, 0]$

- For points in Class 2: $p(y|x) = [0, 1]$

The marginal distribution $p(y) = [0.5, 0.5]$ (2 samples in each class)

KL divergence for each point:

- Class 1 points: $KL([1,0] \,||\, [0.5,0.5]) = 1 \cdot \log(1/0.5) + 0 \cdot \log(0/0.5) = \log(2) \approx 0.693$
- Class 2 points: $KL([0,1] \,||\, [0.5,0.5]) = 0 \cdot \log(0/0.5) + 1 \cdot \log(1/0.5) = \log(2) \approx 0.693$

Average KL divergence = 0.693

Inception Score = exp(0.693) = 2

ii) Calculating FID Score

First, I'll group the real and generated data by class:

Real data:

- Class 1: (0, 0), (0, 1) → Mean $\mu_1 = [0, 0.5]$, Covariance $\Sigma_1 = [[0, 0], [0, 0.5]]$
- Class 2: (1, 0), (1, 1) → Mean $\mu_2 = [1, 0.5]$, Covariance $\Sigma_2 = [[0, 0], [0, 0.5]]$

Generated data:

- Class 1: (0.2, 0.3), (0.1, 0.1) → Mean $\mu_1^g = [0.15, 0.2]$, Covariance $\Sigma_1^g = [[0.005, 0.01], [0.01, 0.02]]$
- Class 2: (0.9, 0.2), (0.7, 0.7) → Mean $\mu_2^g = [0.8, 0.45]$, Covariance $\Sigma_2^g = [[0.02, -0.05], [-0.05, 0.125]]$

FID is calculated as:
$FID = ||\mu - \mu^g||^2 + Tr(\Sigma + \Sigma^g - 2(\Sigma\Sigma^g)^{(1/2)})$

For Class 1:
$||\mu_1 - \mu_1^g||^2 = ||[0, 0.5] - [0.15, 0.2]||^2 = (0-0.15)^2 + (0.5-0.2)^2 = 0.0225 + 0.09 = 0.1125$

For the covariance term, due to the sparsity of Σ_1, I'll make an approximation:
$Tr(\Sigma_1 + \Sigma_1^g - 2(\Sigma_1\Sigma_1^g)^{(1/2)}) \approx Tr(\Sigma_1) + Tr(\Sigma_1^g) \approx 0.5 + (0.005 + 0.02) = 0.525$

$FID_1 \approx 0.1125 + 0.525 = 0.6375$

For Class 2:
$||\mu_2 - \mu_2^g||^2 = ||[1, 0.5] - [0.8, 0.45]||^2 = (1-0.8)^2 + (0.5-0.45)^2 = 0.04 + 0.0025 = 0.0425$

For the covariance term:
$Tr(\Sigma_2 + \Sigma_2^g - 2(\Sigma_2\Sigma_2^g)^{(1/2)}) \approx 0.5 + (0.02 + 0.125) = 0.645$

$FID_2 \approx 0.0425 + 0.645 = 0.6875$

Overall FID = $(FID_1 + FID_2)/2 = (0.6375 + 0.6875)/2 = 0.6625$

Comment on Generated Data Quality

1. The **Inception Score is 2**, which is the maximum possible for a 2-class problem with balanced generation. This indicates good diversity and quality in terms of class coverage.
2. The **FID score is approximately 0.6625**. Lower FID scores indicate better quality, and while this isn't extremely low, it's reasonable considering the small sample size.

Looking at the generated points:

- (0.1, 0.1) and (0.2, 0.3) are good approximations for Class 1
- (0.9, 0.2) is close to (1, 0) from Class 2
- (0.7, 0.7) is reasonably close to (1, 1) from Class 2

The model has generated points that cover both classes with reasonable fidelity to the original distribution. There's a slight shift in the mean values, but the overall structure appears to be captured. The generation quality is acceptable but not perfect, which aligns with the moderate FID score.